



Hyperspectral reconstruction from satellite and airborne observations via a deep hierarchical fusion network across heterogeneous scenarios

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Motivation

Background:

- High-resolution hyperspectral imagery (HR-HSI) is crucial for Earth's surface monitoring but expensive and limited in spatial and temporal coverage.
- Spectral reconstruction through data fusion offers a pathway to generate HR-HSI to address the trade-off between spectral and spatial resolution of current remote sensing imagery.
- Real-world fusion performance, full-spectrum fidelity, and the necessity versus benefit of increasingly complex deep learning models remain insufficiently evaluated.

Research questions:

- How do existing fusion models and our proposed architecture perform and generalize when applied to real, cross-platform remote sensing data?
- How does model complexity affect the accuracy and effectiveness of hyperspectral reconstruction?

Methods

Research areas

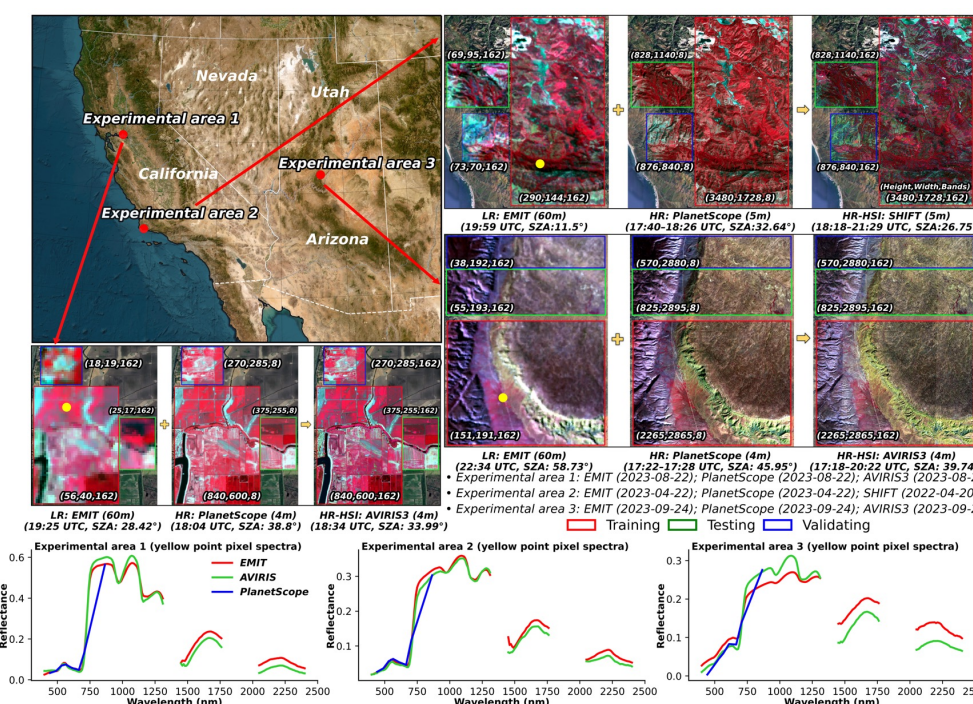


Fig. 1 Overview of the three experimental areas and datasets.

Data Processing

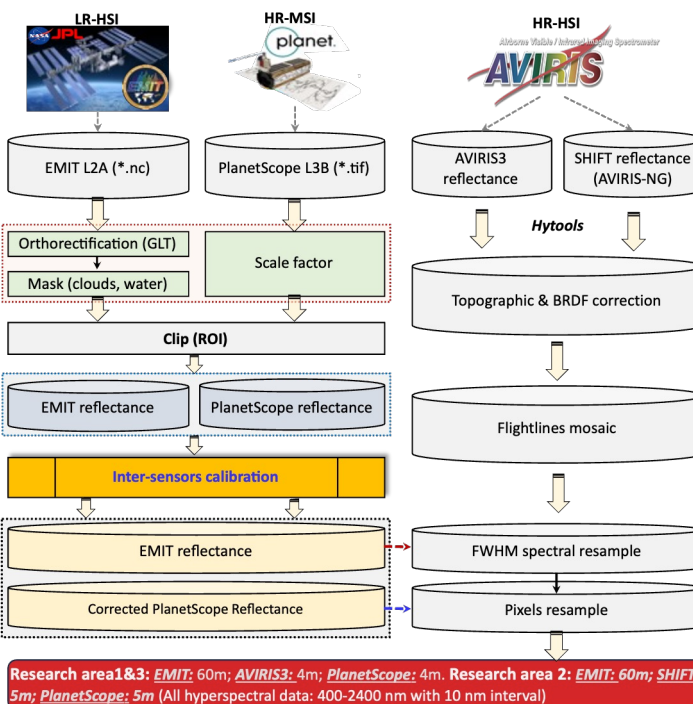


Fig. 2 Summarized workflow for the data pre-processing.

Modeling methods

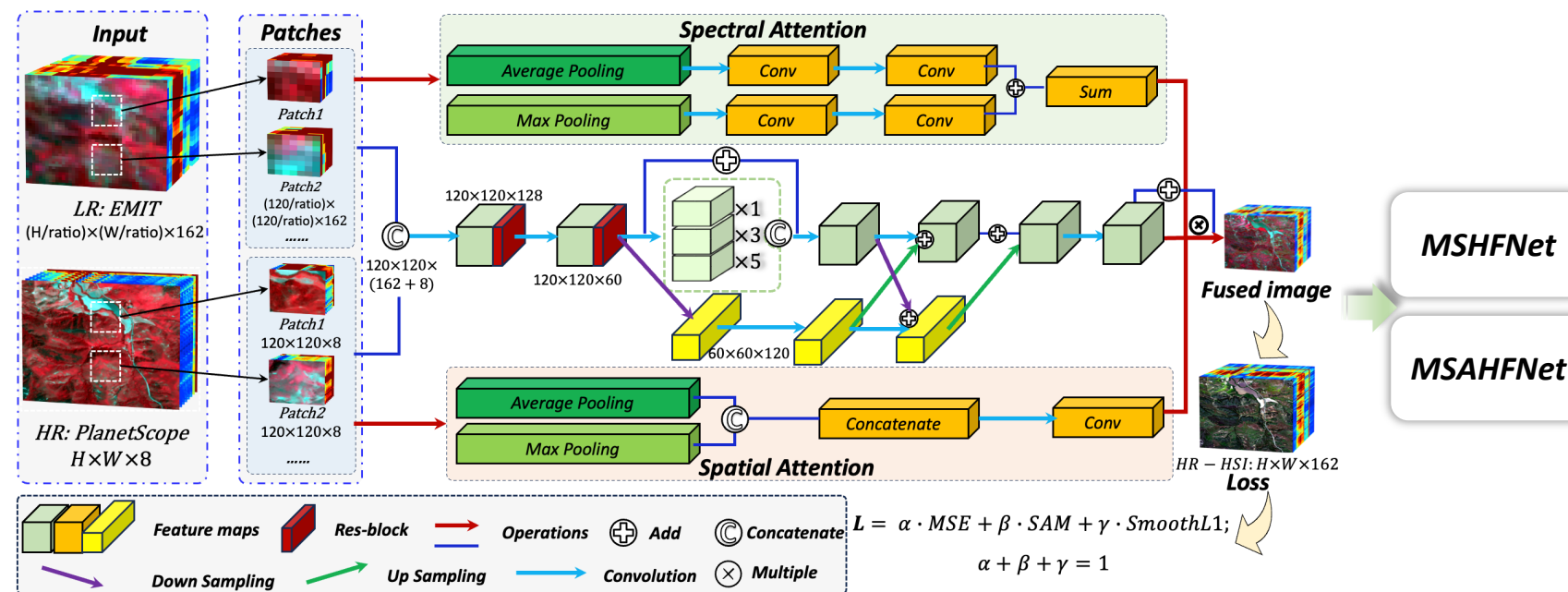


Fig. 3 Architecture of the data fusion framework.

We acknowledge the financial support from NASA Commercial Smallsat Data Scientific Analysis (CSDSA) Program.

Results

Mean quantitative accuracy on test imagery

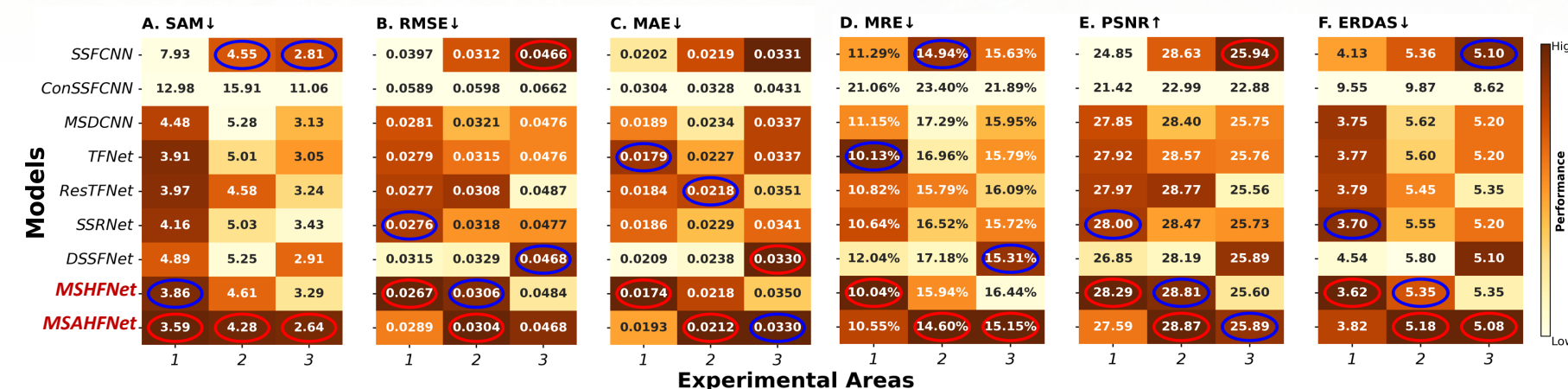


Fig. 4 Quantitative performance of hyperspectral reconstruction models.

- Proposed structure consistently outperform existing models across key evaluation metrics.

Spectral fidelity

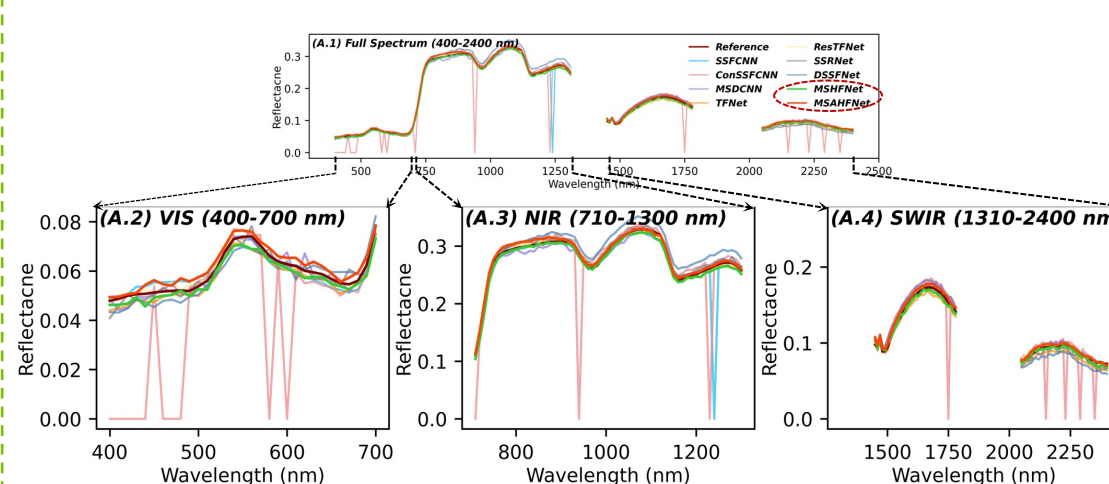


Fig. 5 Spectral comparison between reconstructed and reference hyperspectral reflectance.

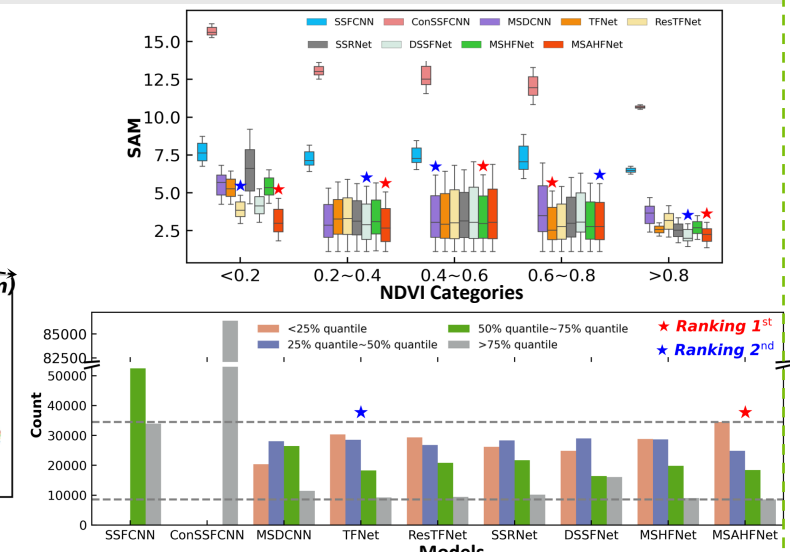


Fig. 6 Pixel-wise spectral fidelity evaluation using SAM.

- Other models showed greater fluctuations and noticeable artifacts.
- Our models have the highest spectral fidelity, showing smoother and more continuous spectral profiles.
- Lowest median SAM values across most NDVI groups and retains the largest proportion of low-SAM pixels.

Spatial fidelity

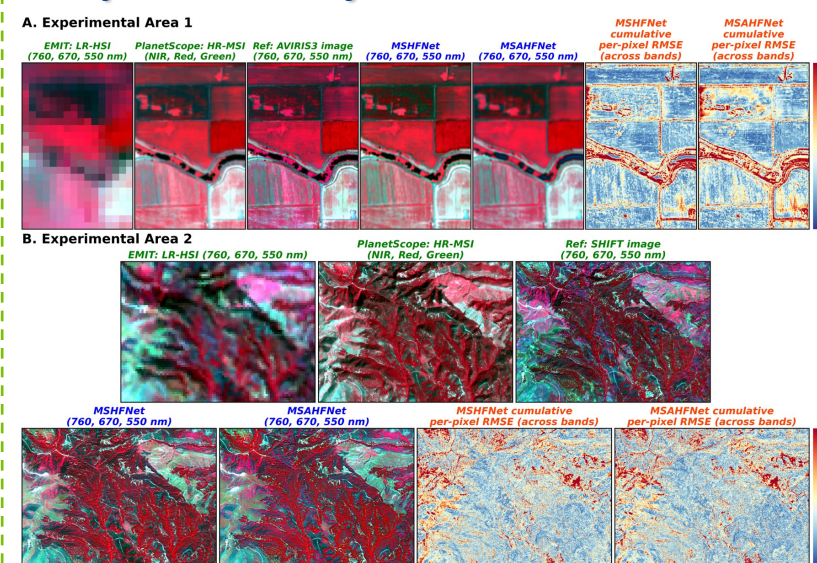


Fig. 9 Visual comparison of reconstructed hyperspectral outputs.

Model complexity and computational efficiency

Models	Model Size (No. of paras in millions)	Training time (hours)		
		Research Area 1	Research Area 2	Research Area 3
SSFCNN	1.14	0.107	1.416	1.881
ConSSFCNN	1.18	0.115	1.46	1.92
MSDCNN	1.85	0.162	2.173	2.929
TFNet	2.5	0.175	1.958	2.459
ResTFNet	2.38	0.172	2.015	2.582
SSRNet	0.71	0.109	1.473	1.93
DSSNet	313.42	1.083	18.895	32.427
MSHFNet	1.49	0.161	2.233	2.617
MSAHFNet	1.50	0.193	2.526	3.036

- Strong capacity to recover detailed spatial information and top performers in spatial fidelity evaluating metrics.
- Our networks achieved a better balance between model complexity, computational cost and reconstruction performance.
- Greater parameter size and model complexity do not necessarily translate to improved hyperspectral reconstruction accuracy.